

Data Warehousing and Business Intelligence

IT3021



Assignment 02

IT23690202

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1. Task 1: Data source for the Assignment

This section describes the data source used for Assignment 2, which is the data warehouse implemented and loaded in Assignment 1. The data warehouse is named DWBI_Superstore and follows a Kimball-style star schema design, optimized for analytical queries and business intelligence operations.

The Superstore dataset contains sales transaction data with customer, product, shipping, and time-related information. In Assignment 1, the data was extracted from multiple source files, loaded into staging tables, transformed into facts and dimensions tables, and stored in a data warehouse. In Assignment 2, this warehouse is used to create an OLAP cube for interactive analysis

1.1 Overview of the Data Warehouse

The DWBI_Superstore database is a subject-oriented, integrated, time-variant, and non-volatile collection of data designed to support decision-making processes related to Super store performance. The central focus is the FactSales table, which stores quantitative measures for each individual Super store.

1.2 Facts and Dimensions

The FactSales table Stores the quantitative measures related to each transaction done in the superstore dataset. Each row in this table represents a single sales record corresponding to an order line in the original dataset. The fact table also contains foreign keys referencing the dimension tables, which allow the fact data to be analyzed from multiple perspectives. following foreign keys are used in factSales table,

DateKey – Reference DimDate

CustomerKey – Reference DimCustomer

ProductKey – Reference DimProduct

ShipModeKey – Reference DimShipMode

During the ETL process, these keys are resolved from the natural keys present in the staging tables using lookup transformations.

1.2.2 Date Dimension

represents the time dimension of the warehouse. It contains detailed attributes related to dates derived from the OrderDate field in the dataset. The table includes attributes such as Year, Month, Quarter, Day which enable time-based analysis such as identifying sales trends across months, quarters, or years.

1.2.3 Product Dimension

stores product-related descriptive attributes including ProductID, Category, SubCategory, and ProductName. This dimension enables analysis of sales performance by product category, sub-category, or individual products. It allows business users to identify best-selling products, evaluate product category performance, and analyze profitability by product groups.

1.2.4 Customer Dimension

Contains descriptive information about customers. It includes attributes such as CustomerID, CustomerName, Segment, Country, City, State, PostalCode, and Region. This is selected as the Slowly Changing Dimension Type 2, because customer profile and geographical attributes may change over time and such changes should be preserved historically. It keeps track of changes in customer information over time by maintaining historical records. The attributes EffectiveStartDate, EffectiveEndDate, and IsCurrent are used to manage versioning.

1.2.5 Ship Mode Dimension

contains the different shipping methods used for order delivery. This dimension includes attributes related to shipping modes such as Standard Class, Second Class, First Class, and Same Day.

1.3 Assumptions

The grain of the fact table is one row per sales transaction record from the original Superstore dataset.

- The grain of the FactSales table is defined as one record per order line transaction. Each row represents a single product sold in an order. The fact table stores the measurable sales metrics such as Sales, Quantity, Discount, and Profit, while referencing the related dimension keys including DateKey, CustomerKey, ProductKey, and ShipModeKey.

1.4 Star Schema for Superstore warehouse (Facts and dimensions)

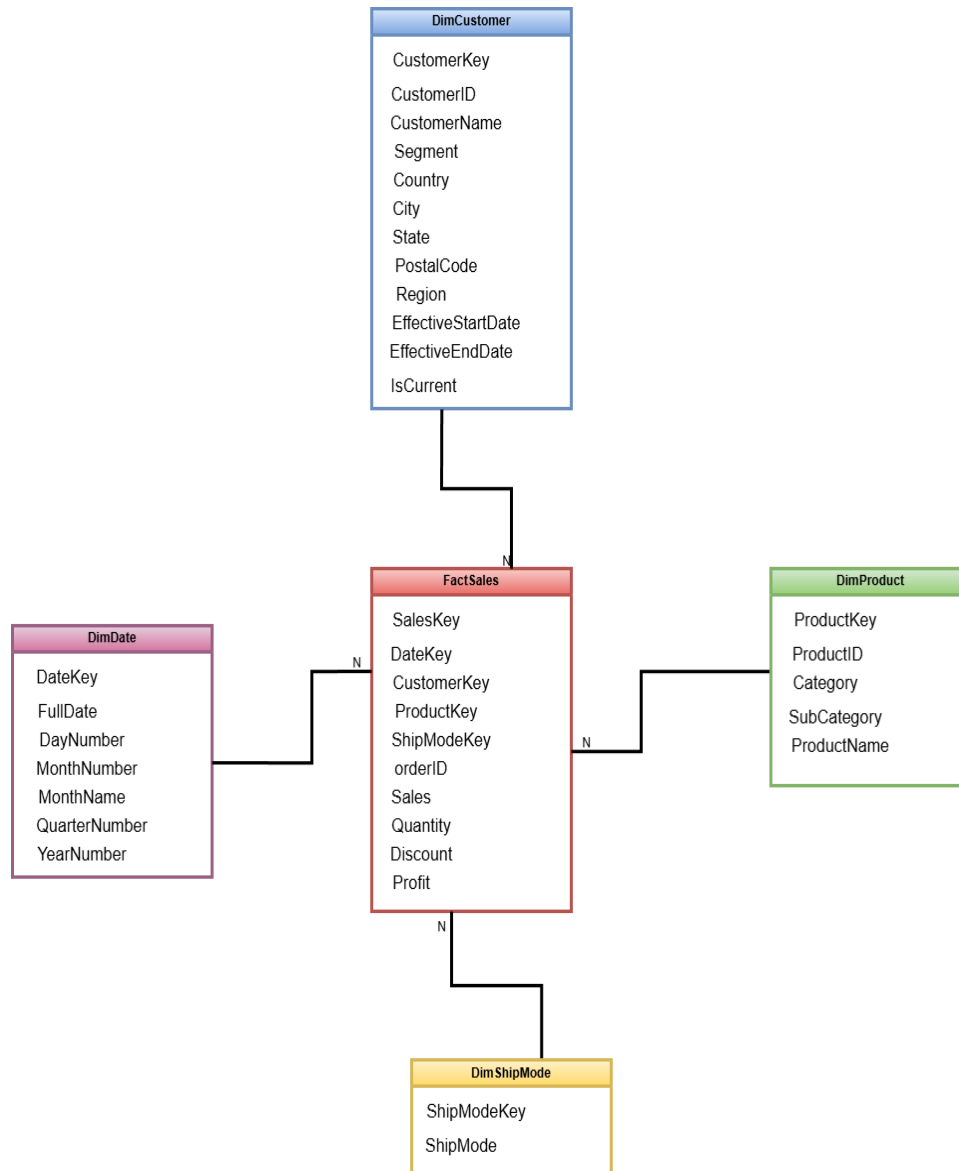


Figure 1. Star Schema of the Superstore

The dimensional model follows a star schema where FactSales is the central fact table and four-dimension table is connected to it through surrogate keys as DimDate, DimCustomer, DimProduct and DimShipMode. The design follows Kimball's dimensional modelling methodology, optimizing for analytical query performance through denormalization and surrogate key lookups.

2. Task 2: SSAS Cube implementation

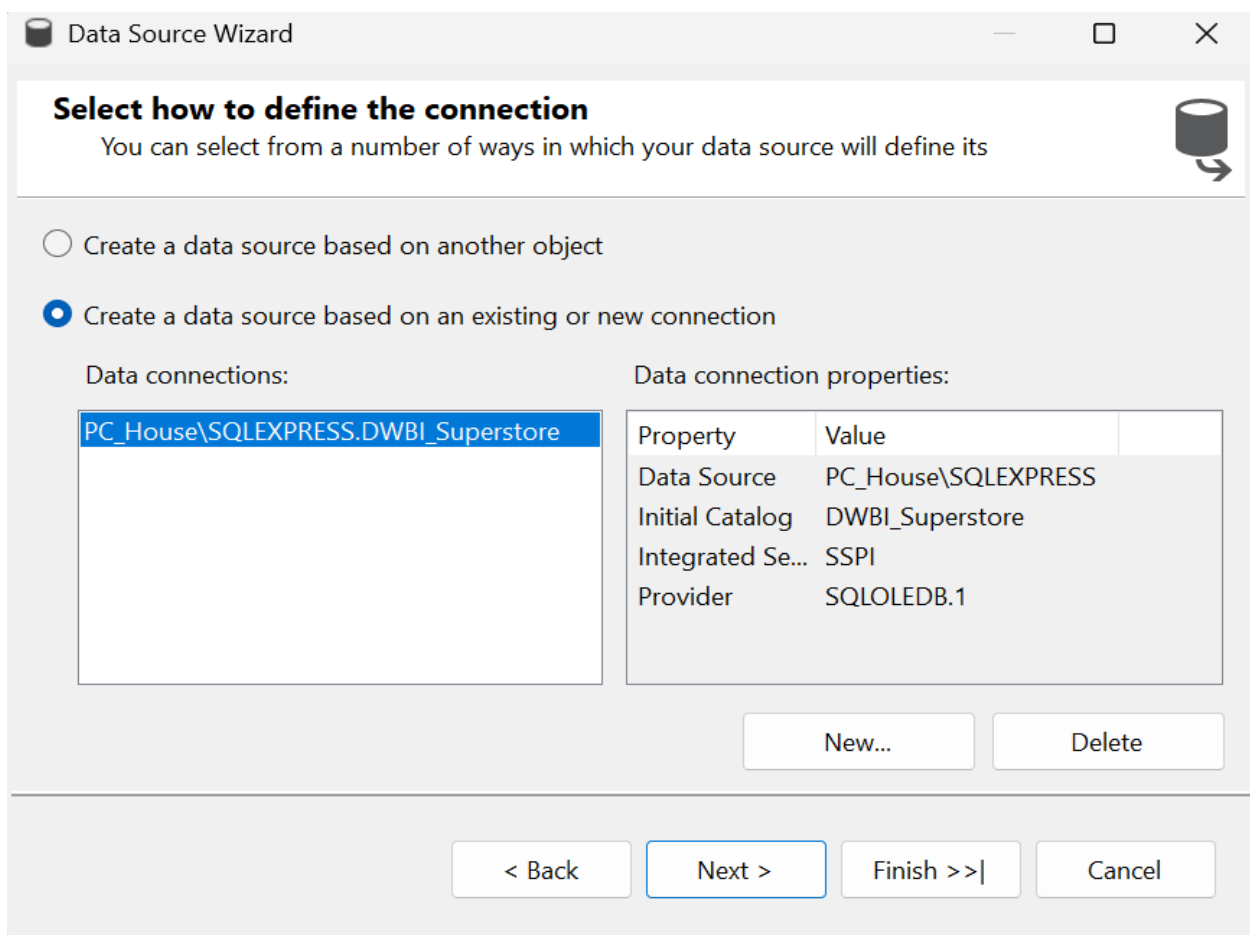
2.1 Analysis Services Project Environment

The OLAP cube was developed within a SQL Server Analysis Services Multidimensional Project using SQL Server Data Tools (SSDT) in Visual Studio. This development environment provides the necessary components required to define data connections, dimensions, cube structures, and processing configurations.

2.2 Creating Data Source (DS)

As the first step, in the Data Source Wizard, “**Create a data source based on an existing or new connection**”. The connection was created to the DWBI_Superstore database, which contains the fact and dimension tables populated during the ETL process implemented in Assignment 1.

The data source allows the cube to retrieve data directly from the warehouse tables and ensures that the cube remains synchronized with the underlying relational schema



Data Source Wizard

Select how to define the connection
You can select from a number of ways in which your data source will define its

☐ Create a data source based on another object

☒ Create a data source based on an existing or new connection

Data connections:

Property	Value
Data Source	PC_House\SQLEXPRESS
Initial Catalog	DWBI_Superstore
Integrated Se...	SSPI
Provider	SQLOLEDB.1

New... Delete

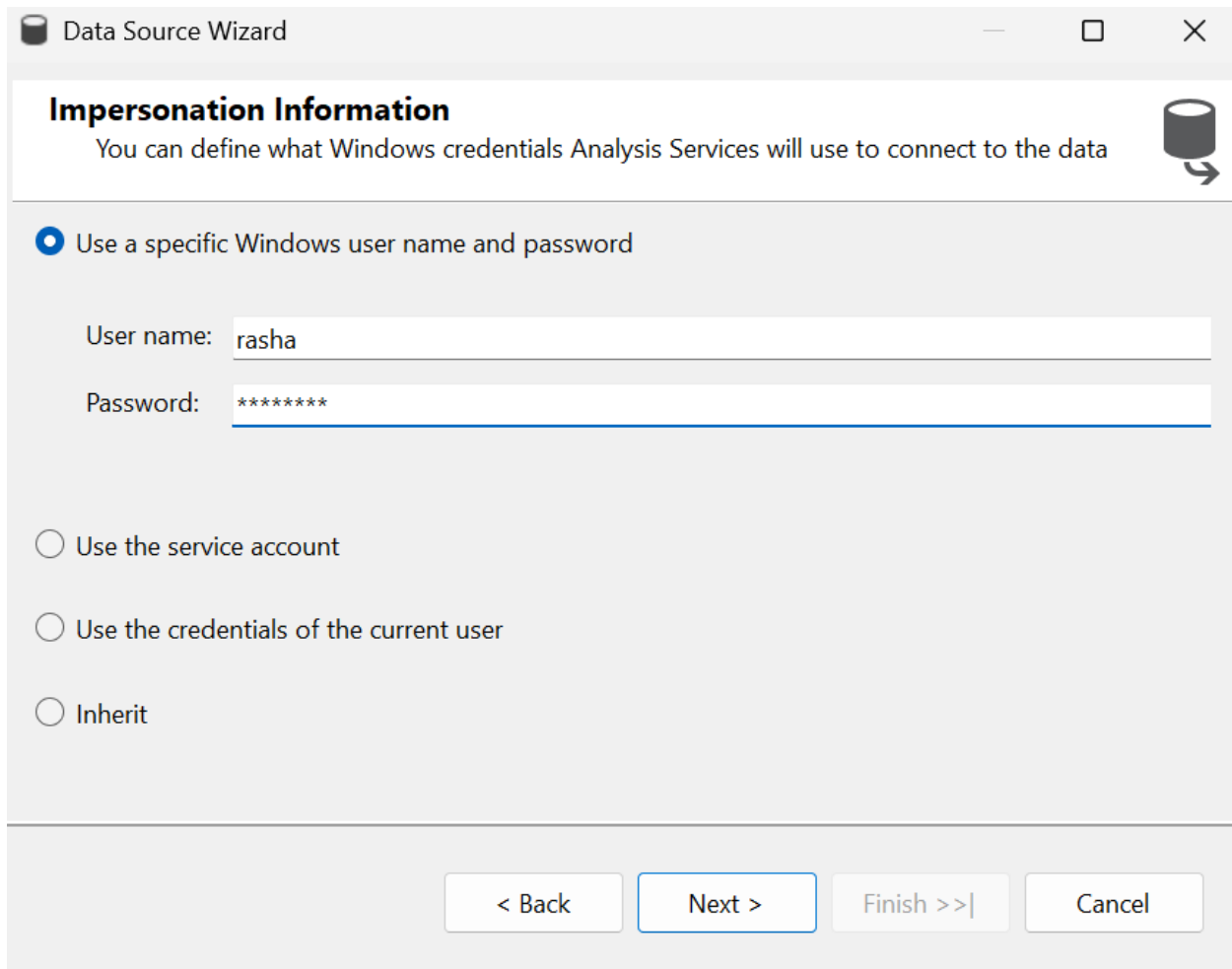
< Back Next > Finish >>| Cancel

Figure 2. New Data Source Creation

Again, in the Data Source Wizard, the **Impersonation Information** which is used to authenticate the user to access the data source, The “**Use a specific Windows username and password**” option was selected.

Use the Username and password in your laptop.

After this point a name for the data source is assigned and the data source is connected successfully.



The screenshot shows a Windows-style dialog box titled "Data Source Wizard". The main heading is "Impersonation Information" with a subtitle "You can define what Windows credentials Analysis Services will use to connect to the data". There is a database icon with a refresh arrow in the top right corner. The first option, "Use a specific Windows user name and password", is selected with a blue radio button. Below this, there are two text input fields: "User name:" containing the text "rasha" and "Password:" containing seven asterisks "*****". Below these fields are three unselected radio button options: "Use the service account", "Use the credentials of the current user", and "Inherit". At the bottom of the dialog, there are four buttons: "< Back", "Next >" (which is highlighted with a blue border), "Finish >>|", and "Cancel".

Figure 3. Impersonation Information Configuration

2.3 Creating Data Source View (DSV)

A Data Source View (DSV) was created to define the logical representation of the data warehouse schema used by the cube. The Data Source View acts as an abstraction layer between the relational database and the OLAP cube. All fact tables and dimensions tables are included.

The Data Source View visually represents the relationships between the fact table and the dimension tables. These relationships are based on surrogate keys defined in the dimensional model.

A data source view is created using the **Data Source View Wizard** the previously created data source is selected and then the Fact table, FactSales, is selected. After selecting the Fact table the **Add Related Table** is used to include the connected Dimension tables.

Data Source View Wizard

Select Tables and Views
Select objects from the relational database to be included in the data source

Available objects:

Name	Type
OLE DB Destination (dbo)	Table
orders_source (dbo)	Table
Stg_Customer (dbo)	Table
Stg_Orders (dbo)	Table
Stg_Product (dbo)	Table
vw_SuperstoreCombined (d...	View
SalesCompletion (dbo)	Table

Included objects:

Name	Type
FactSales (dbo)	Table
DimCustomer (dbo)	Table
DimDate (dbo)	Table
DimProduct (dbo)	Table
DimShipMode (dbo)	Table

Filter:

☐ Show system objects

Add Related Tables

< Back **Next >** **Finish >>|** **Cancel**

Figure 4. New Data Source View Creation

After selecting the relevant tables, **DWBI_Superstore** is assigned as the name for the data source. After creating the data source view, the table diagram can be viewed in the design view.

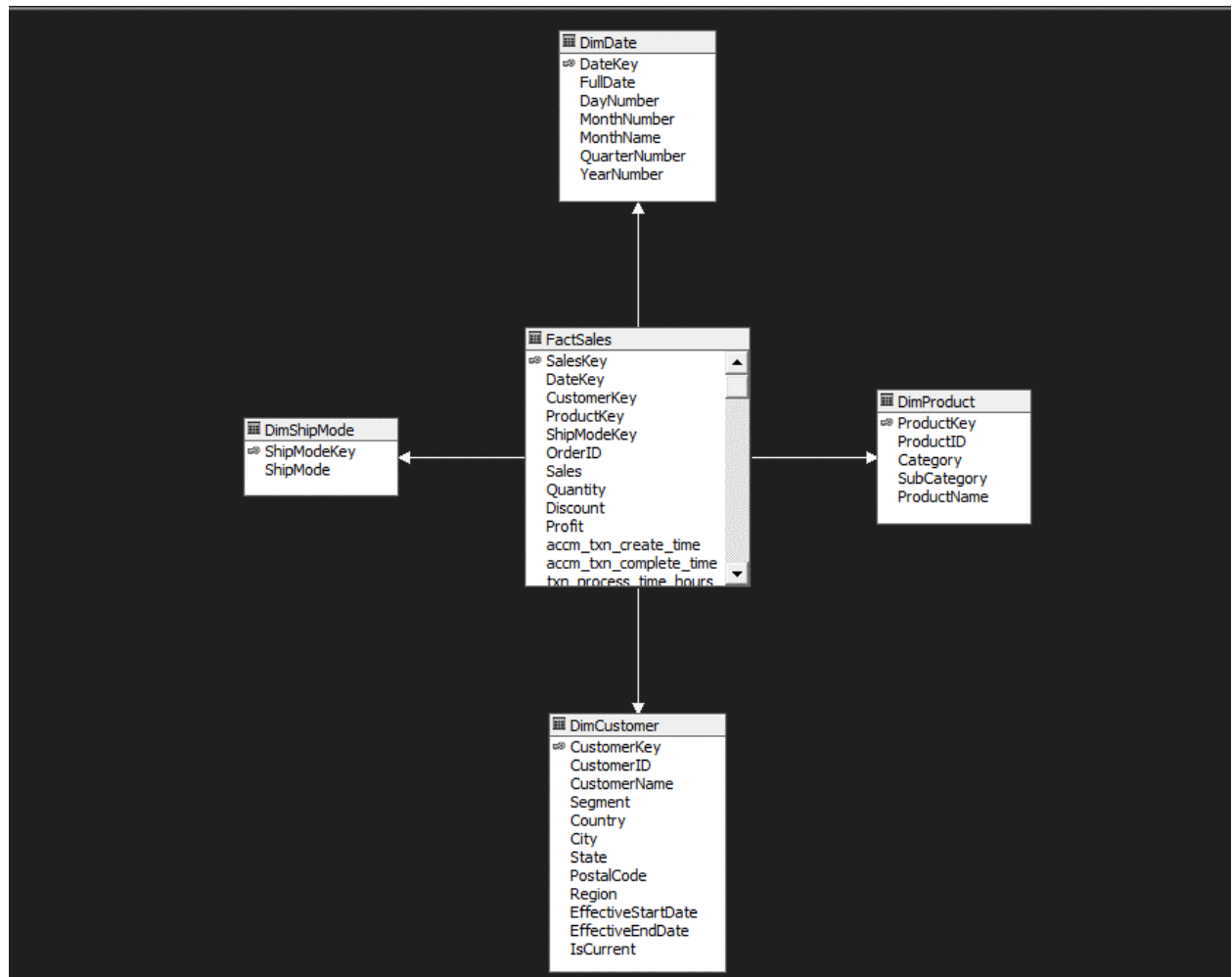


Figure 5. Data Source View Schema

As shown in the diagram, all the dimension tables are connected correctly to FactSales through the respective foreign keys.

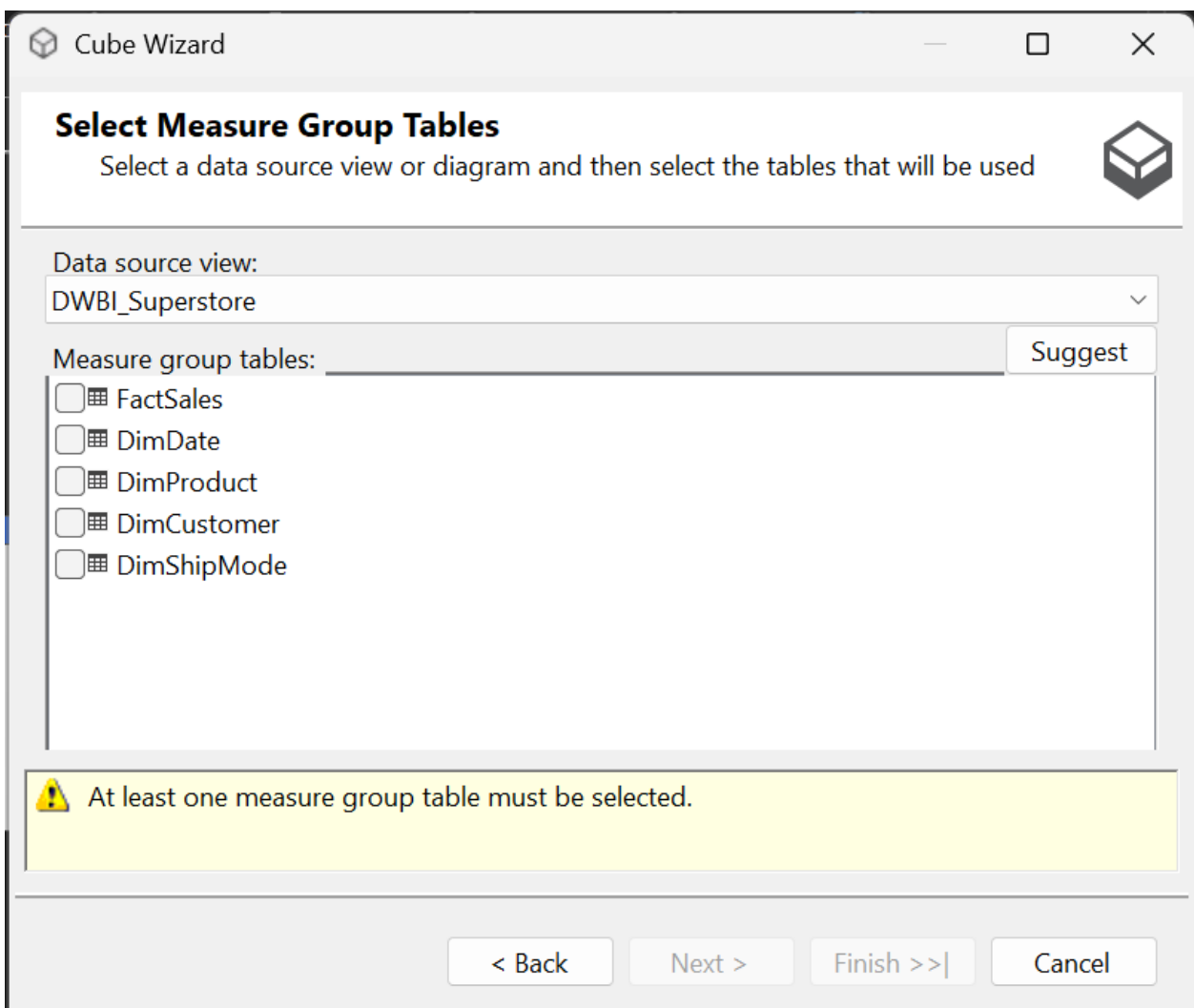
2.4 Cube Structure

Defining the fact table and dimensions, the **OLAP cube** was created by combining these elements into a multidimensional model.

A cube is created using the **Cube Wizard**. After selecting the Use existing tables option in the starting page, the data source view, DWBI_Superstore created in the previous step is selected.

In the Select Measure Group Tables the FactSales table is chosen as it is the only table containing the measures in the data warehouse.

The cube structure enables users to perform multidimensional analysis by combining measures with various dimension attributes.



The screenshot shows the 'Cube Wizard' window with the title bar 'Cube Wizard'. The main heading is 'Select Measure Group Tables' with a subtitle 'Select a data source view or diagram and then select the tables that will be used'. A cube icon is in the top right. Below the subtitle, 'Data source view:' is followed by a dropdown menu showing 'DWBI_Superstore'. Under 'Measure group tables:', there is a list of tables with checkboxes: FactSales, DimDate, DimProduct, DimCustomer, and DimShipMode. A 'Suggest' button is to the right of the list. At the bottom, a yellow warning box contains a triangle icon and the text 'At least one measure group table must be selected.' The footer contains four buttons: '< Back', 'Next >', 'Finish >>|', and 'Cancel'.

Figure 6. SSAS Cube Fact Table Selection

After that, the next Step is to select existing dimensions as shown.

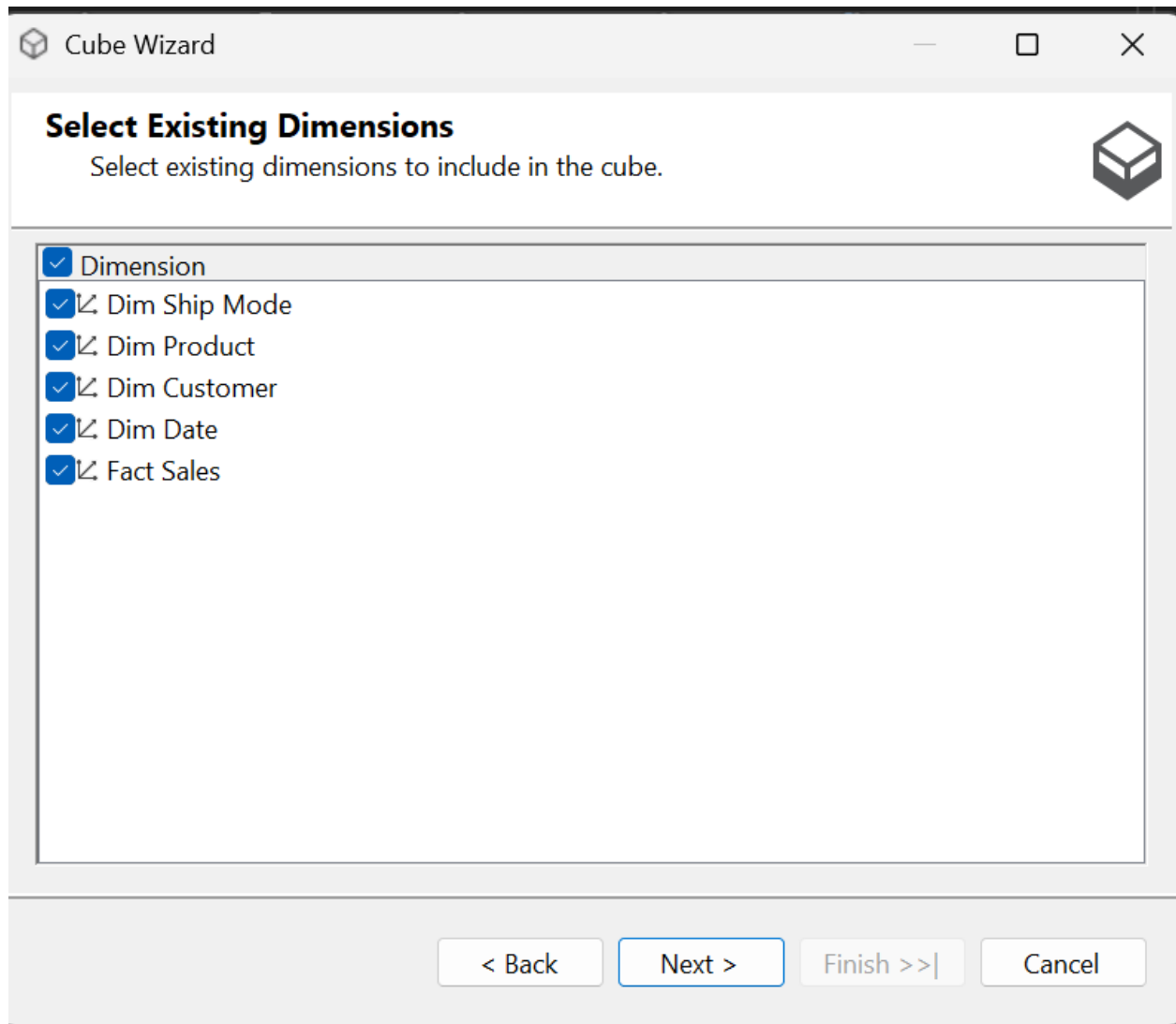


Figure 7. SSAS Cube Dimension Selection

After selecting the dimension table, a name is assigned to the cube as “DWBI_Superstore.cube”

The Star Schema built as a cube can be viewed in the Cube Design view.

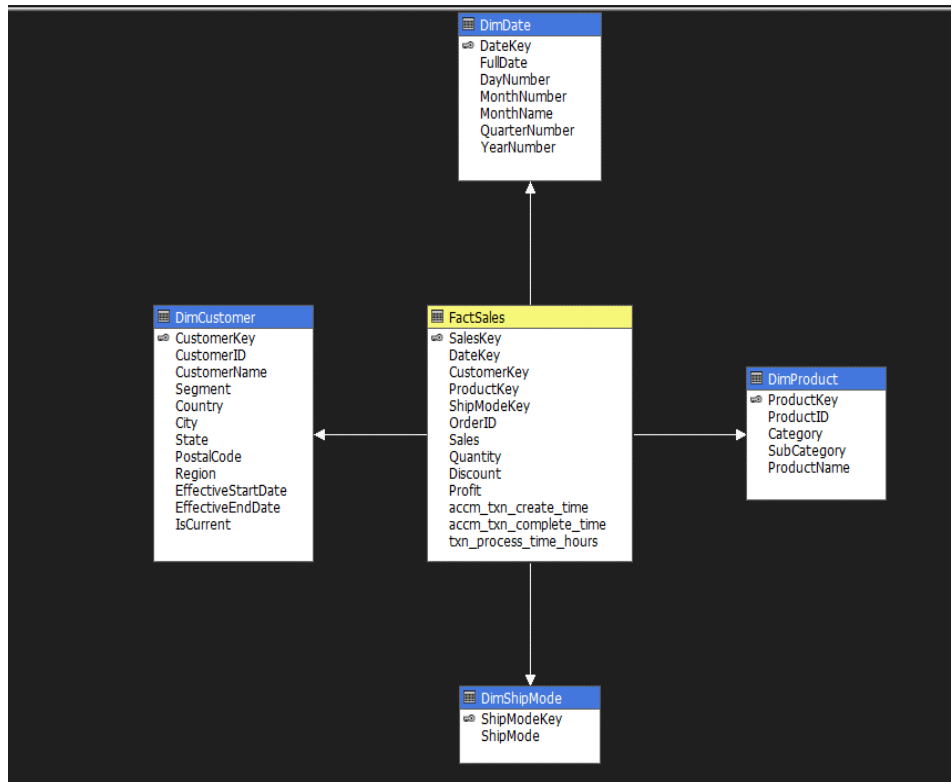


Figure 8. SSAS Cube Schema

In the dimensions section, only the key attributes of the dimensions are shown. The other attributes should be added manually using the Edit Dimension panel. A completed example is shown below for DimCustomer.

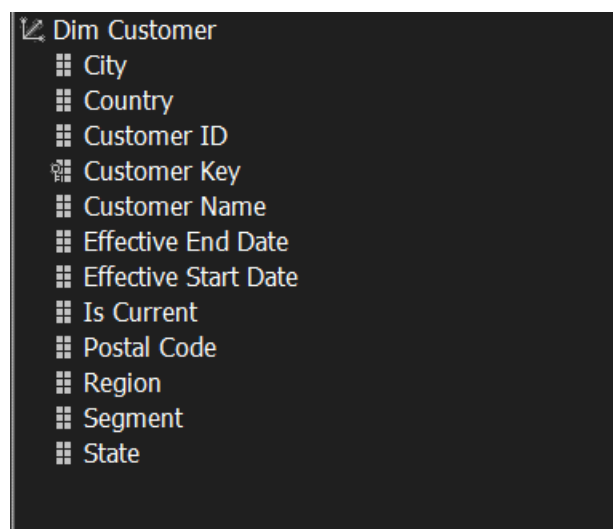


Figure 9. DimGuest Dimension Attributes

2.5 Implementing hierarchy in the cube

Hierarchies organize attributes into levels, enabling users to move between summarized and detailed views of data. The hierarchy was named as **Hierarchy** as it is

The following were defined in the cube:

Time Hierarchy

- Year Number → Quarter Number → Month Name → Day Number

This hierarchy allows users to explore sales trends across different time levels.

It improves usability by allowing users to easily perform drill-down and roll-up operations during analysis.

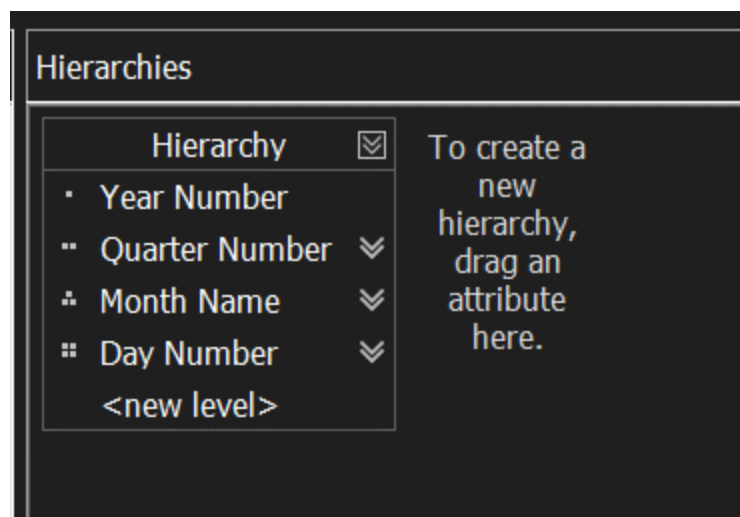


Figure 10. DimDate Attribute hierarchy



Figure 11. DimDate Attribute Relationship

2.6 Cube Deployment and Processing

Once the cube structure was completed, the cube was deployed and processed in SSAS. Processing the cube loads the warehouse data into the OLAP engine and prepares it for analytical querying. After successful processing, the cube became available for connection from reporting tools. Successful processing shows that the cube is ready for browsing and analytical queries.

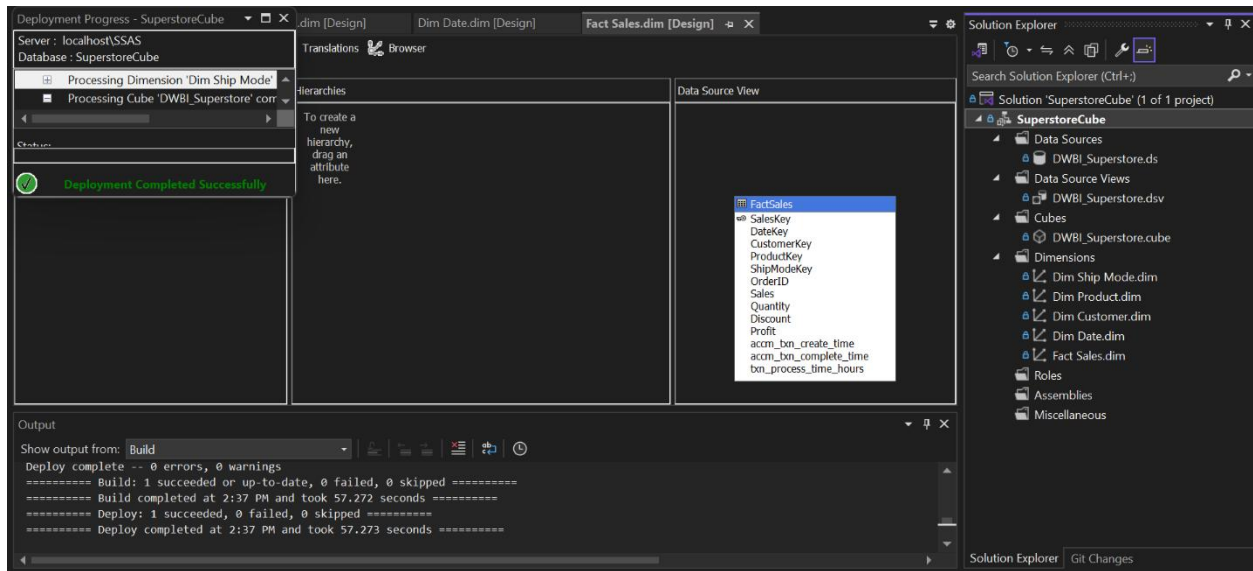


Figure 12. final Method Deployment Successfully

3. Task 3: Demonstration of OLAP operations

3.1 Connecting Excel to the OLAP Cube

To perform analytical operations on the cube, Microsoft Excel was connected to the SuperstoreCube created in SSAS. Excel provides built-in OLAP capabilities through PivotTables and Pivot Charts, allowing users to interactively explore multidimensional data.

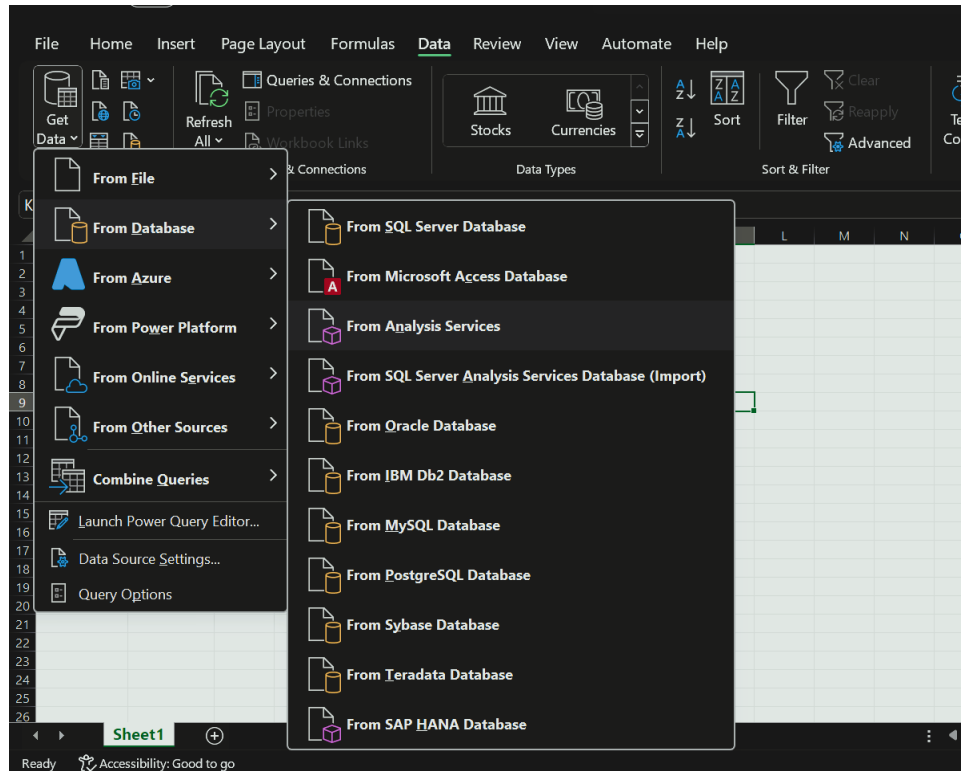


Figure 13. Excel Connection Method

1. Microsoft Excel was opened, and a new workbook was created.
2. The **Data** tab was selected from the Excel ribbon.
3. The **Get Data** option was chosen, and the connection was established using **From Database** → **From Analysis Services**.
4. The server's name containing the SSAS cube was entered.
5. The **SuperstoreCube** database and cube were selected.
6. The data was imported into Excel as a **PivotTable Report**.

After establishing the connection, Excel displayed the cube fields including measures and dimensions. These fields were used to create PivotTables and Pivot Charts for analytical exploration.

3.2 OLAP operations Using Pivot Tables

3.2.1 Roll-up Operation

The **roll-up operation** aggregates detailed data into higher levels of a hierarchy.

In this analysis, the Roll-Up operation aggregates detailed data into higher levels of the hierarchy. In this example, monthly sales data was summarized at the Year level using the Date hierarchy.



Figure 14. Roll-Up Operation Excel Visualization

Steps:

- Click inside the PivotTable
- In the field list on the right, expand **DimDate** to choose the **Hierarchy**
- Drag **Hierarchy** to the **Rows** area
- Drag **Sales** (measure) to the **Values** area

The table will show data at the Year level by default, and this aggregated view is the rolled-up view.

3.2.2 Drill-Down Operation

The **drill-down operation** allows users to navigate from summarized data to more detailed levels within a hierarchy.

In this analysis, the Drill-Down operation allows users to navigate from summarized data to more detailed levels. In this example, yearly sales data was expanded into Quarter and Month levels.

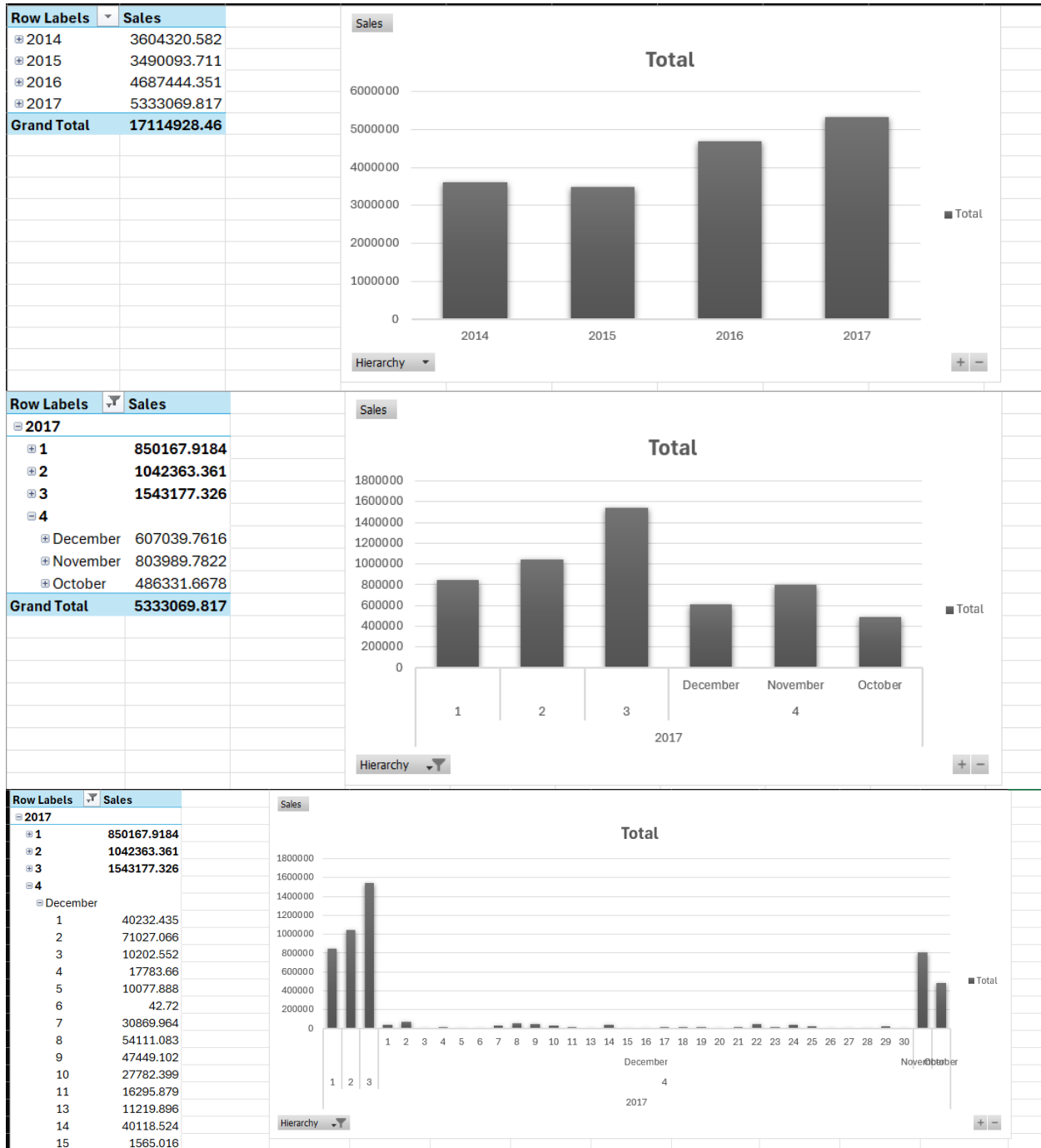


Figure 15. Drill-Down Operation Excel Visualization

3.2.3 Slice Operation

The slice operation filters the data by selecting a single dimension value.

In this analysis, slice operation filters the cube by selecting a single dimension value. In this example, the Region slicer was used to filter data for the West and central region.

Slice operation was demonstrated by filtering the cube data for a single dimension value, such as Region = West. This allowed the analysis to focus on one specific subset of the data while keeping other dimensions unchanged.

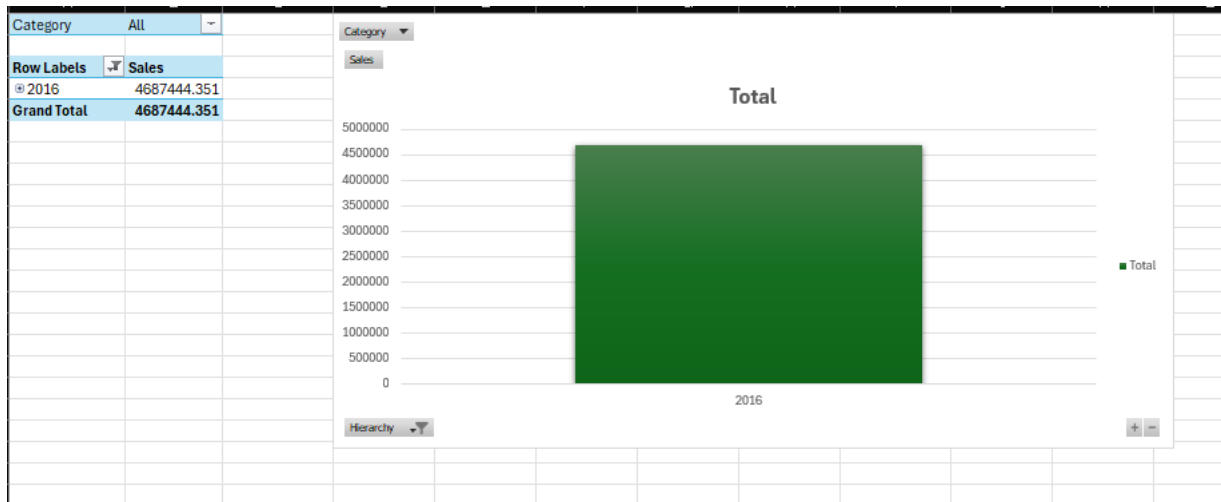


Figure 16. Slice Operation Excel Visualization

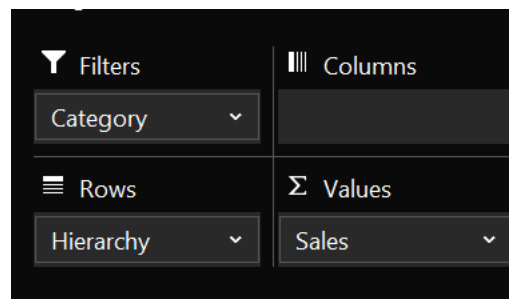


Figure 17. Slice Operation PivotTable Configuration

- Create a PivotTable
- Drag **Hierarchy** into the Rows section
- Drag **Sales** into the Values section
- Drag **Category** into the Filter section

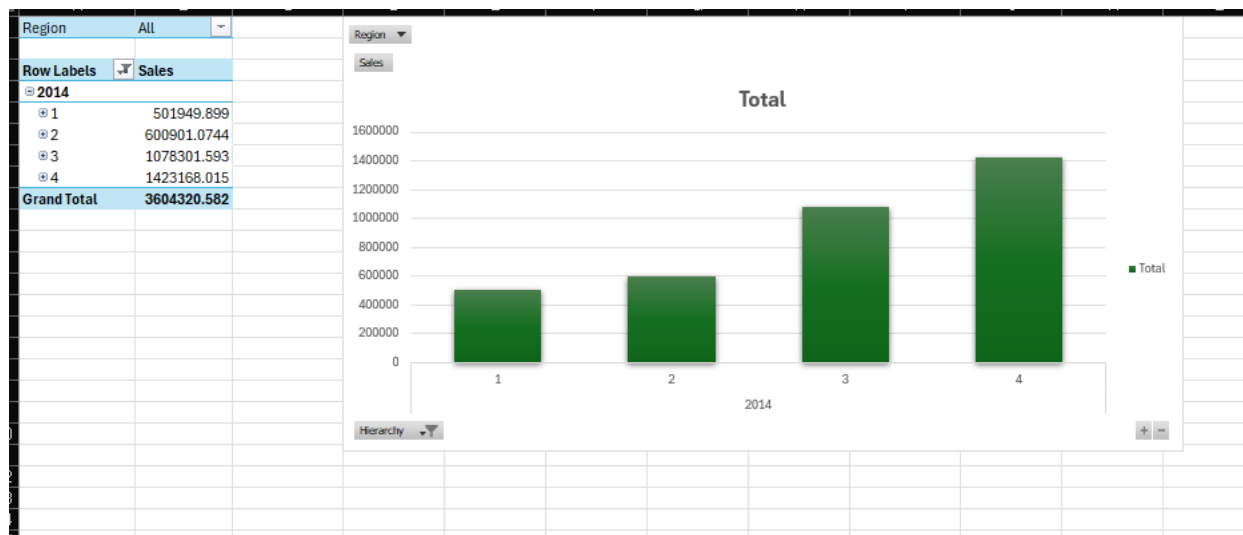


Figure 18. Slice Operation Excel Visualization 1

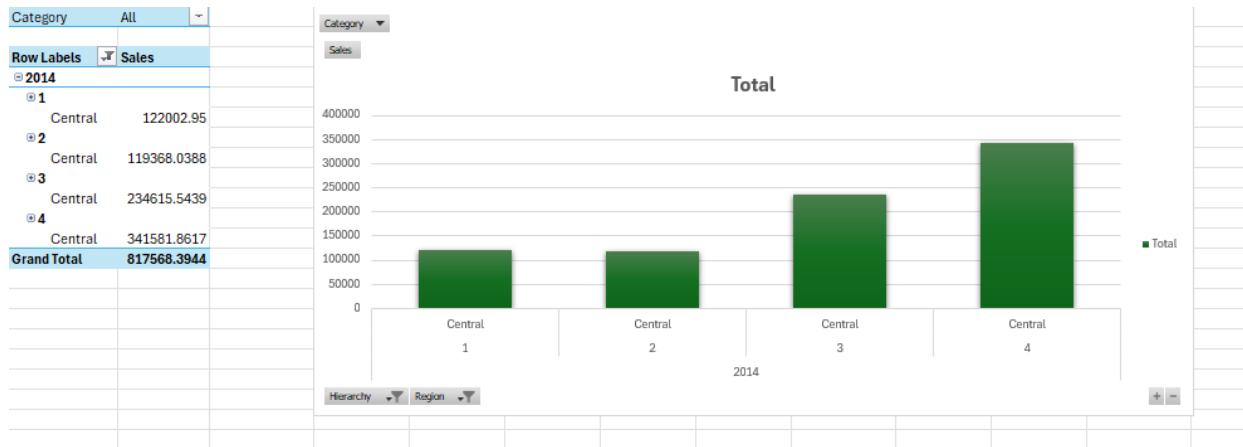


Figure 19. Slice Operation Excel Visualization 2

3.2.4 Dice Operation

The dice operation filters data using multiple dimensions simultaneously.

In this implementation, two slicers were used:

- Region
- Category

Selecting a specific region and category filters the data to display only the records matching both conditions.

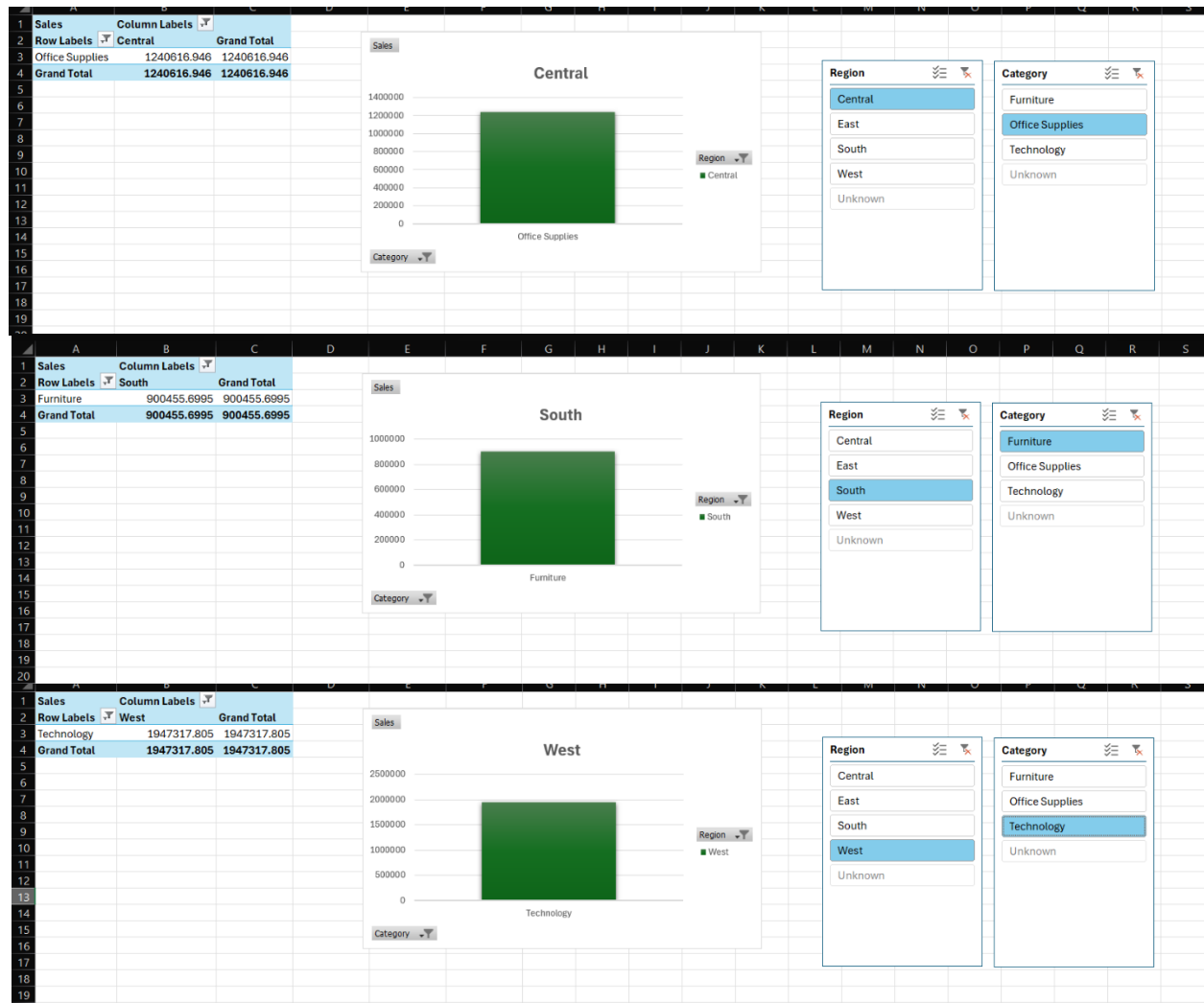


Figure 20. Dice Operation Excel Visualization

3.2.5 Pivot Operation

The **pivot operation** changes the orientation of data to view it from different perspectives.

In Excel PivotTables, pivoting is achieved by moving fields between the Rows, Columns, and Filters areas. This allows the same dataset to be analyzed in multiple ways without modifying the underlying data.

For example, product categories can be displayed across regions or regions can be displayed across product categories.

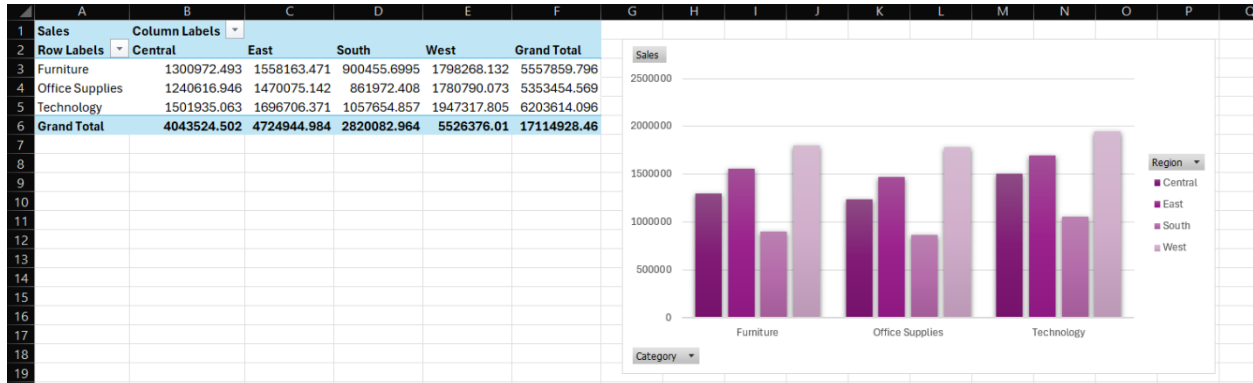


Figure 21. Before Pivot Operation Excel Visualization

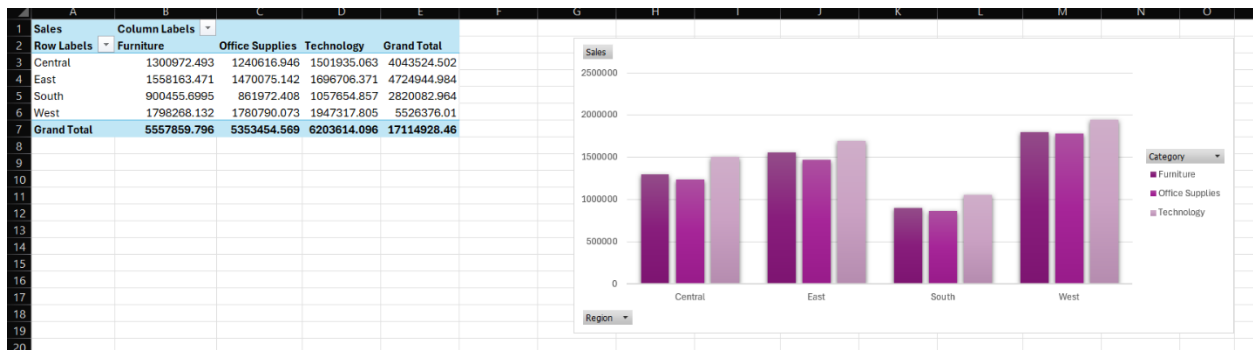


Figure 22. After Pivot Operation Excel Visualization

Steps:

- Create a PivotTable
- First, Put Category (from DimProduct) in Rows area, Region (from DimCustomer) in Columns area and add Sales to the Values area.
- Then switch the Rows and Columns, put Category in Columns area and Region in the Rows area

4. Task 4: Power BI Reports

The Power BI reports were developed using the **DWBI_Superstore cube**, which was previously created using SQL Server Analysis Services (SSAS). The cube contains a fact table and several dimension tables that enable multidimensional analysis.

Power BI Desktop was connected to the cube using a Live Connection to Analysis Services. This approach allows the reports to directly query the cube instead of importing the data into Power BI. Because of this, the measures defined in the cube can be reused directly within the Power BI visuals.

4.1. Report 01 – Matrix Visualization

To create the matrix visual, the Category and Sub-Category fields from the Dim Product table were added to the Rows section. This created a hierarchy that allows users to expand and collapse product categories to view the corresponding sub-categories.

The matrix visual enables users to perform detailed comparisons and analyze sales performance across different dimensions simultaneously. It also supports hierarchical exploration by allowing users to expand or collapse category levels for more detailed analysis.

Category	Central	East	South	West	Total
▢ Furniture	1,300,972.49	1,558,163.47	900,455.70	1,798,268.13	5,557,859.80
Bookcases	241,005.48	251,012.77	137,871.56	262,468.51	892,358.32
Chairs	568,144.01	685,173.14	371,540.19	805,055.62	2,429,912.96
Furnishings	162,247.43	190,994.27	121,049.62	235,250.89	709,542.21
Tables	329,575.58	430,983.29	269,994.34	495,493.11	1,526,046.31
▢ Office Supplies	1,240,616.95	1,470,075.14	861,972.41	1,780,790.07	5,353,454.57
Appliances	179,609.08	205,043.53	115,672.80	256,632.79	756,958.21
Art	46,246.95	53,786.54	32,451.59	59,809.78	192,294.86
Binders	367,780.10	391,873.32	289,694.72	534,248.72	1,583,596.86
Envelopes	30,379.24	35,694.64	18,768.32	37,783.19	122,625.38
Fasteners	5,143.24	5,956.18	3,640.82	7,800.38	22,540.63
Labels	19,477.41	19,998.89	15,864.65	29,164.60	84,505.56
Paper	147,977.28	176,581.61	105,744.30	190,702.87	621,006.06
Storage	363,930.54	472,831.39	252,374.48	513,768.48	1,602,904.89
Supplies	80,073.11	108,309.03	27,760.72	150,879.26	367,022.13
▢ Technology	1,501,935.06	1,696,706.37	1,057,654.86	1,947,317.80	6,203,614.10
Accessories	332,284.07	388,380.52	243,176.67	418,711.30	1,382,552.55
Copiers	255,016.73	298,515.76	146,607.84	245,866.07	946,006.40
Machines	272,182.93	360,488.00	231,001.30	437,692.98	1,301,365.20
Phones	642,451.34	649,322.09	436,869.05	845,047.46	2,573,689.94
Total	4,043,524.50	4,724,944.98	2,820,082.96	5,526,376.01	17,114,928.46

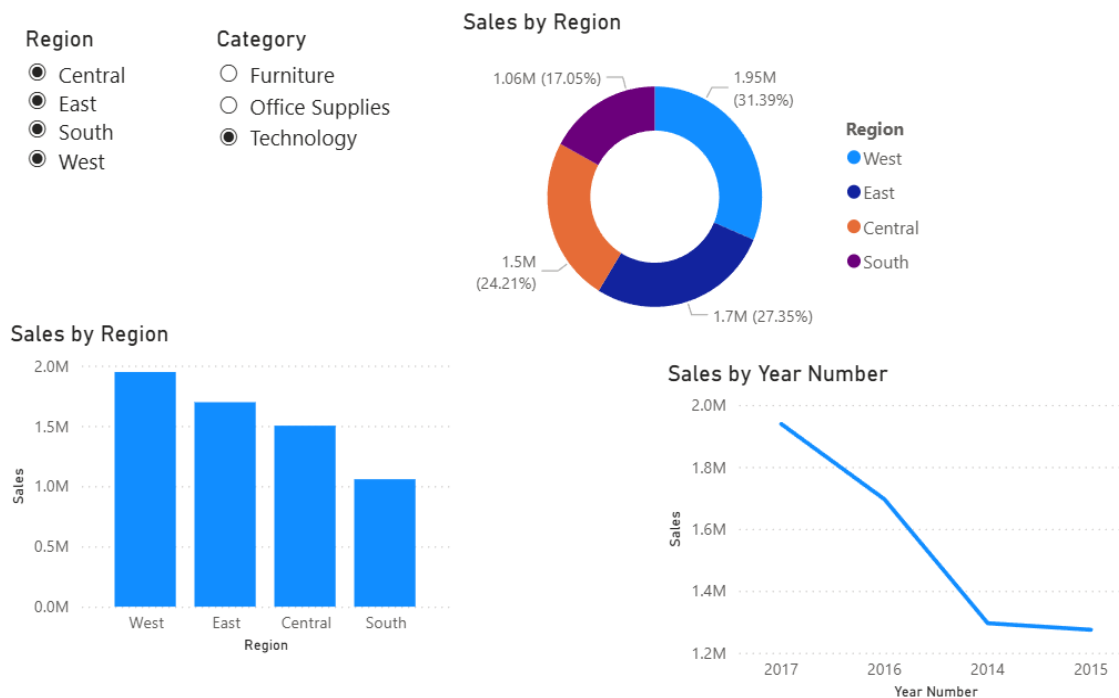
Figure 23. Power BI Matrix Visualization

4.2 Report 02 – Cascading Slicers with Multiple Charts

Report 2 implements interactive filtering using multiple slicers to allow users to dynamically explore the dataset. Two slicers were created using Region and Category. Selecting a value in these slicers automatically filters all visuals in the report page.

- A **column chart** showing Sales by product category
- A **pie chart** showing the distribution of Sales by region
- A **donut chart** showing the proportion of Sales across different categories
- A **line chart** showing the trend of Sales over time using the Year attribute

These visuals work together with the slicers to provide an interactive dashboard where users can filter the data and analyze sales performance from multiple perspectives.



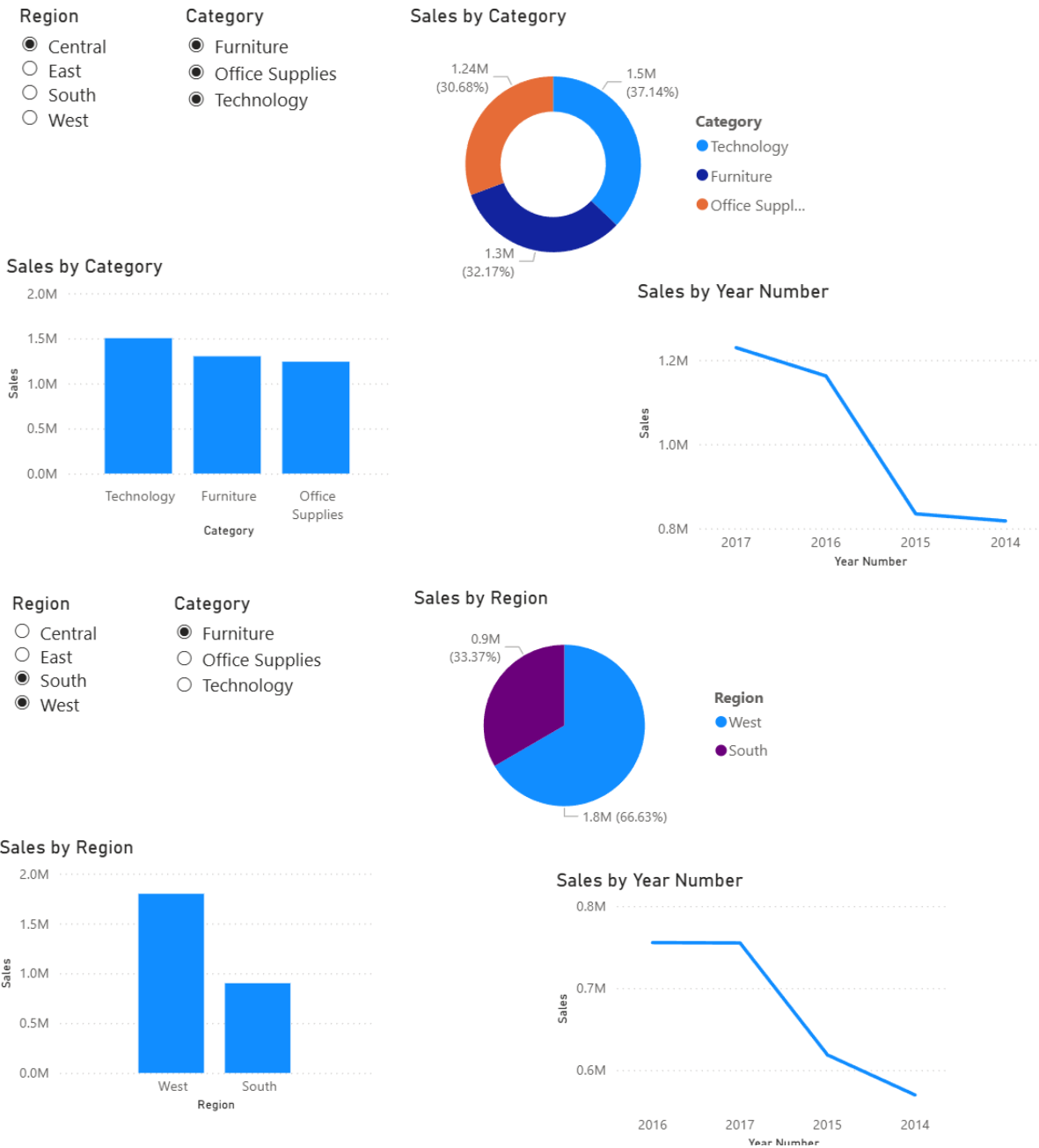


Figure 24. Power BI Cascading Slicer Visualization

4.3 Report 03 – Drill-Down Hierarchical Report

Report 3 demonstrates the Drill Down functionality, which allows users to analyze data at different levels of detail. A column chart was used to display sales data across a time hierarchy.

The hierarchy was created using Year, Quarter and Month attributes from the Dim Date table. Initially, the chart displays total sales aggregated by year. When the drill-down option is enabled, users can click on a specific year to view sales by quarter, and then further drill down into monthly sales.

This feature allows users to explore sales trends over time and identify patterns or changes in sales performance at different levels of the time hierarchy.

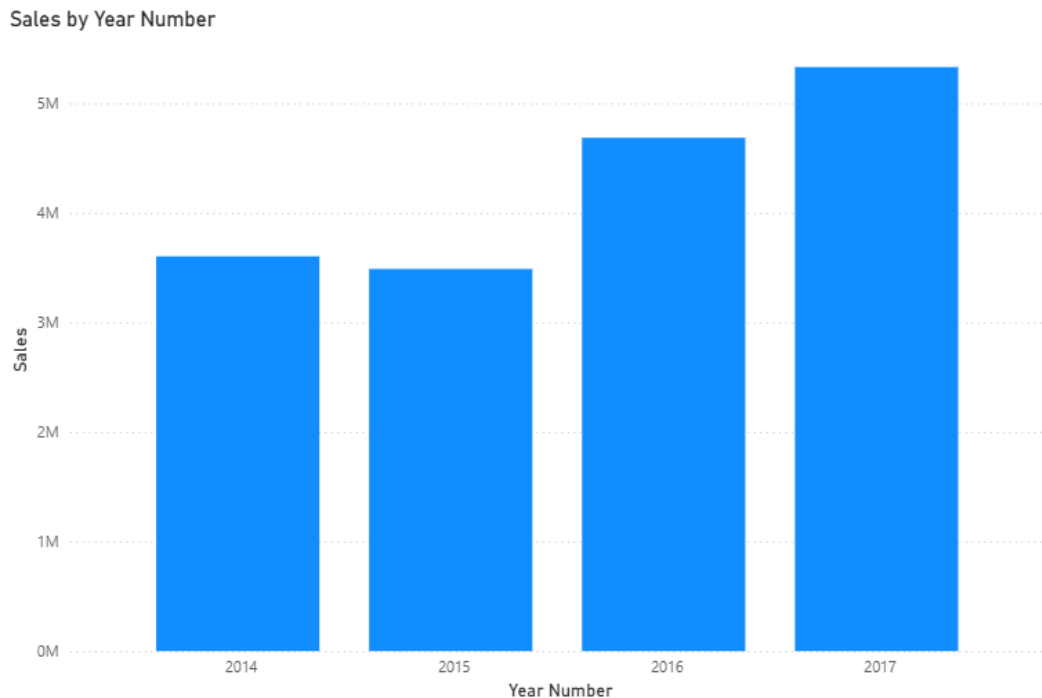


Figure 25.. Power BI Drill Down Visualization (Year)

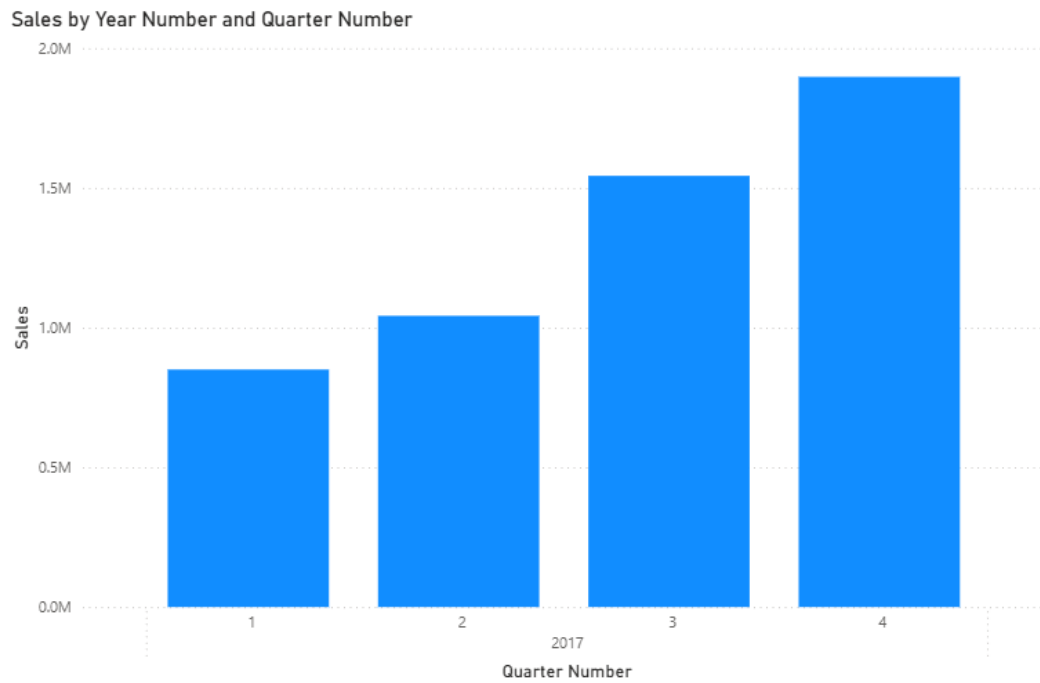


Figure 26. Power BI Drill Down Visualization (Quarter)

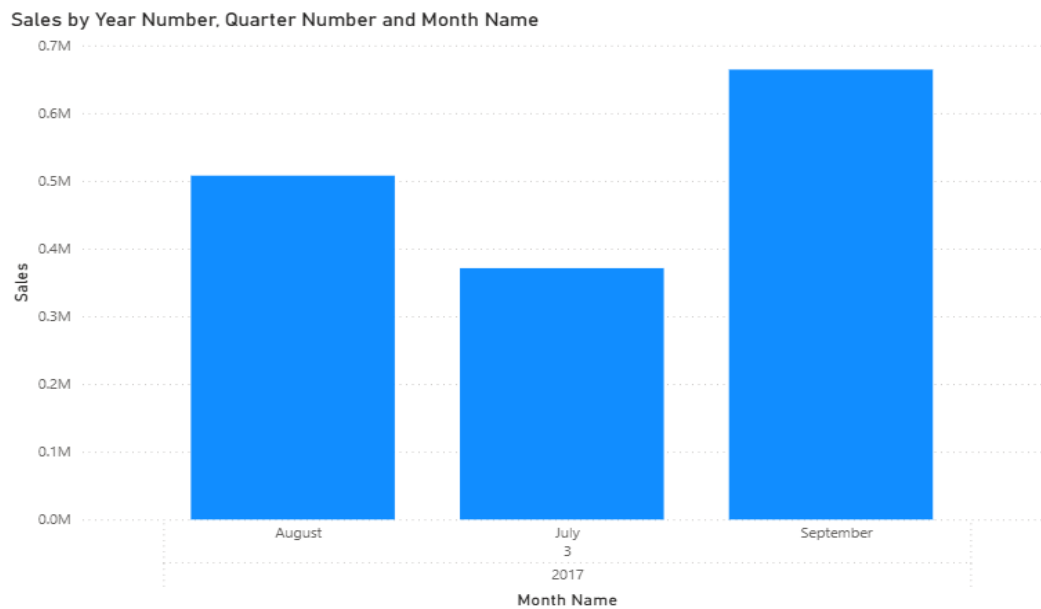


Figure 27. Power BI Drill Down Visualization (Month)

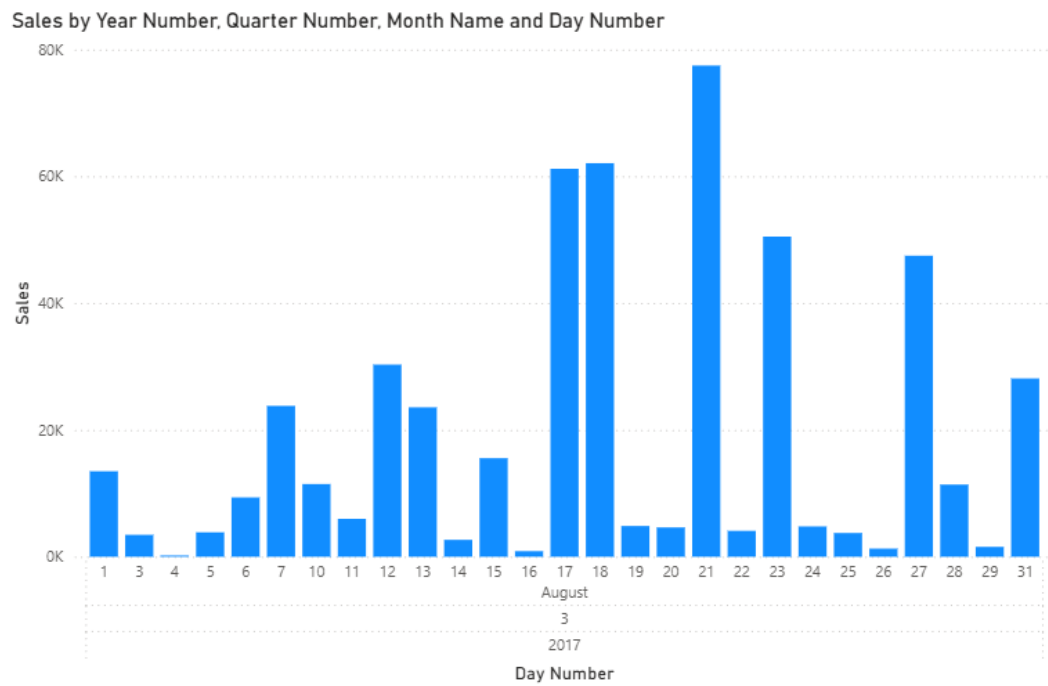


Figure 28. Power BI Drill Down Visualization (Day)

4.4 Report 04 – Drill-Through Report

Report 4 demonstrates the Drill Through functionality, which allows users to navigate from a summarized view of the data to a more detailed view.

A separate report page named Product Details was created for the drill-through report. The Category field from the Dim Product table was added as the drill-through filter. This allows the page to receive context from the main report page.

When users right-click on a category value in the main report and select Drill Through → Product Details, the report navigates to the Product Details page. This page contains a table visual displaying detailed information including Product Name, Sales, Profit and Quantity for the selected category. A Back button was added to allow users to return to the main report page.

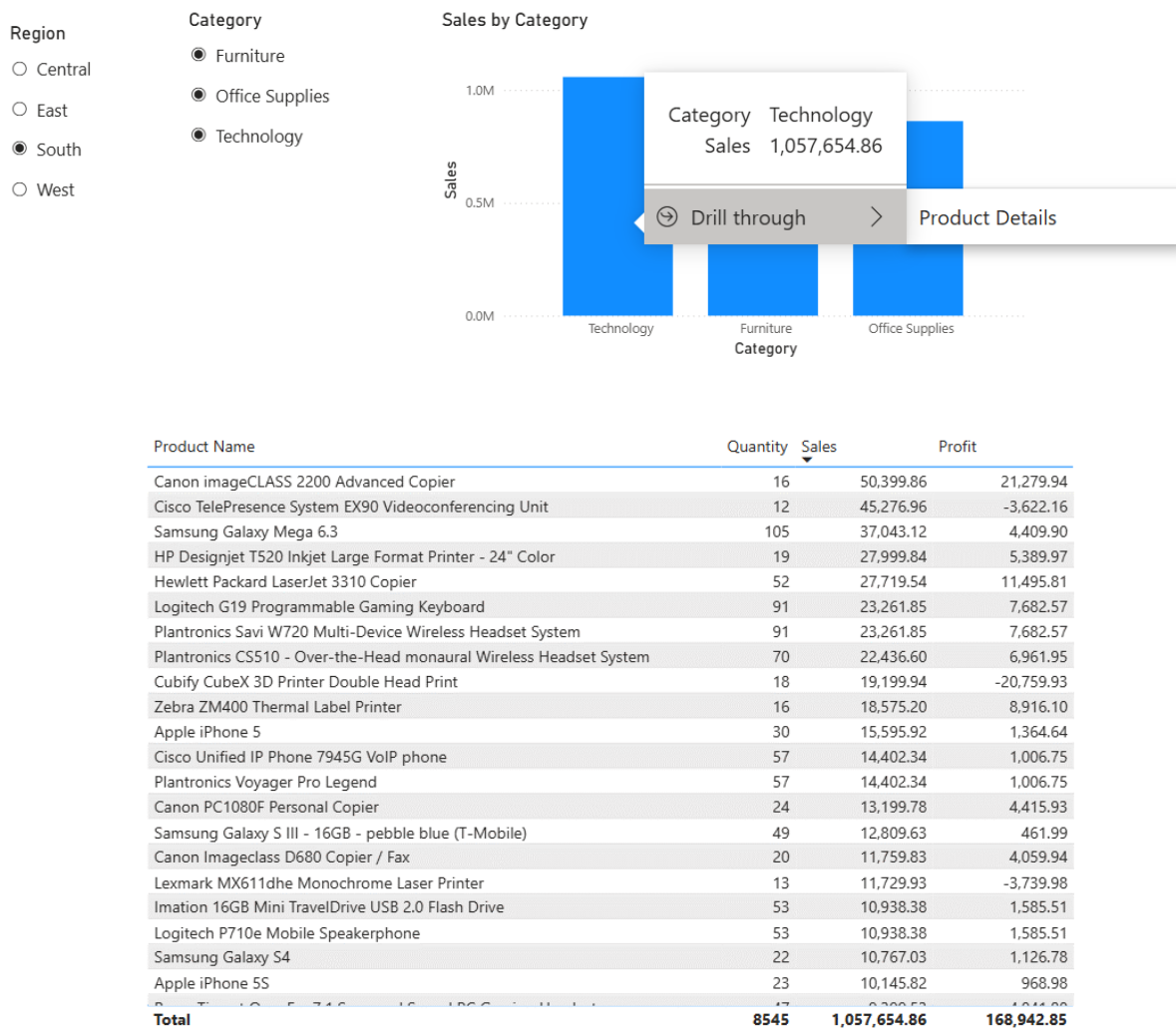


Figure 29. Power BI Drill-Through Visualization